

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June 2021 Worked Solutions

May/June 2021 paper rebuilt with the worked solution after each question.

9

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 May/June 2021

May/June 2021 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Modelling with Sequences & Series

1. Adina is saving money to buy a new computer. She saves £5 in week 1, £5.25 in week 2, £5.50 in week 3 and so on until she has enough money, in total, to buy the computer.

She decides to model her savings using either an arithmetic series or a geometric series.

Using the information given,

- (a) (i) state with a reason whether an arithmetic series or a geometric series should be used,
- (ii) write down an expression, in terms of n , for the amount, in pounds (£), saved in week n .

(3)

Given that the computer Adina wants to buy costs £350

- (b) find the number of weeks it will take for Adina to save enough money to buy the computer.

(4)

Worked Solution - Question 1

Topic group

1. Choose the model

The weekly amounts are **5**, **5.25**, **5.50**, and so on. The increase is a constant **0.25**, so an arithmetic series should be used.

2. Write the n th amount

The first amount is **5** and the common difference is **0.25**, so the amount saved in week n is $5 + 0.25(n - 1)$.

3. Set up the total

After n weeks, $S_n = \frac{n}{2}\{2(5) + (n - 1)(0.25)\}$.

4. Form the inequality

Adina needs $S_n \geq 350$, so $\frac{n}{2}(9.75 + 0.25n) \geq 350$.

5. Solve for n

This is $n^2 + 39n - 2800 \geq 0$. The positive root is **36.89...**, so the first whole number that works is $n = 37$.

6. Check the week

After **36** weeks the total is below **350**, and after **37** weeks it is **351.50**, so it takes **37** weeks.

Final answer

(a)(i) Arithmetic series.

(a)(ii) $5 + 0.25(n - 1)$.

(b) **37** weeks.

WMA12/01 MAY/JUNE 2021

8 marks

Question 2

Laws of Logarithms

2.

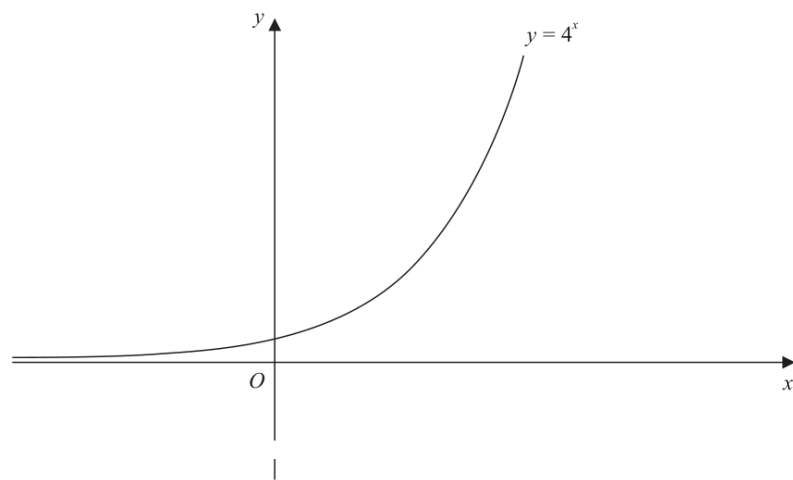


Figure 1

Figure 1 shows a sketch of the curve with equation $y = 4^x$

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(a) On Diagram 1, sketch the curve with equation

(i) $y = 2^x$

(ii) $y = 4^x - 6$

Label clearly the coordinates of any points of intersection with the coordinate axes.

(4)

The curve with equation $y = 2^x$ meets the curve with equation $y = 4^x - 6$ at the point P .

(b) Using algebra, find the exact coordinates of P .

(4)

Question 2 continued

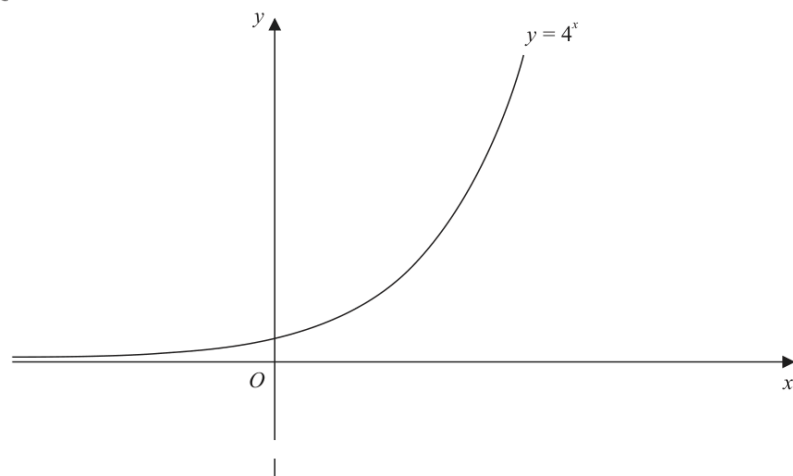


Diagram 1

Worked Solution - Question 2

Topic group

1. Describe y equals two to the x

$y = 2^x$ is an increasing exponential curve. It crosses the y -axis at $(0, 1)$ and has no x -axis intercept.

2. Describe y equals four to the x minus six

$y = 4^x - 6$ is the graph of $y = 4^x$ translated 6 units down. It crosses the y -axis at $(0, -5)$.

3. Find the x -axis intercept

For $y = 4^x - 6$, set $4^x - 6 = 0$. Hence $4^x = 6$ and $x = \log_4 6$.

4. Set up the intersection equation

At P , $2^x = 4^x - 6$.

5. Use a substitution

Let $u = 2^x$. Then $4^x = (2^x)^2 = u^2$, so $u = u^2 - 6$.

6. Solve for u

$u^2 - u - 6 = 0$, so $(u - 3)(u + 2) = 0$. Since $u = 2^x > 0$, $u = 3$.

7. State P

$2^x = 3$, so $x = \log_2 3$, and $y = 3$. Therefore $P = (\log_2 3, 3)$.

Final answer

(a)(i) Sketch $y = 2^x$ through $(0, 1)$ with no x -axis intercept.

(a)(ii) Sketch $y = 4^x - 6$ through $(0, -5)$ and $(\log_4 6, 0)$.

(b) $P = (\log_2 3, 3)$.

Question 3

Proof

3. (i) Prove that for all single digit prime numbers, p ,

$$p^3 + p \text{ is a multiple of } 10 \quad (2)$$

- (ii) Show, using algebra, that for $n \in \mathbb{N}$

$$(n + 1)^3 - n^3 \text{ is not a multiple of } 3 \quad (3)$$

Worked Solution - Question 3

Topic group

1. List the cases

The single digit prime numbers are **2, 3, 5**, and **7**.

2. Check each prime

$$2^3 + 2 = 10, 3^3 + 3 = 30, 5^3 + 5 = 130, \text{ and } 7^3 + 7 = 350.$$

3. Conclude part one

Each result is a multiple of **10**, so $p^3 + p$ is a multiple of **10** for all single digit primes.

4. Expand the difference of cubes

$$(n + 1)^3 - n^3 = (n^3 + 3n^2 + 3n + 1) - n^3 = 3n^2 + 3n + 1.$$

5. Factor the multiple of three

$$3n^2 + 3n + 1 = 3n(n + 1) + 1.$$

6. Use the remainder

$3n(n + 1)$ is a multiple of **3**, so $3n(n + 1) + 1$ is one more than a multiple of **3**. Therefore it is not a multiple of **3**.

Final answer

(i) Proven by checking $p = 2, 3, 5, 7$.

(ii) Proven: $(n + 1)^3 - n^3$ leaves remainder **1** on division by **3**.

Question 4

Binomial Expansion

4. (a) Find, in ascending powers of x , up to and including the term in x^3 , the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

- (b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

(3)

Without calculating 2.0125^{13}

- (c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

Worked Solution - Question 4

Topic group

1. Write the binomial terms

$$\left(2 + \frac{x}{8}\right)^{13} = 2^{13} + \binom{13}{1}2^{12}\frac{x}{8} + \binom{13}{2}2^{11}\left(\frac{x}{8}\right)^2 + \binom{13}{3}2^{10}\left(\frac{x}{8}\right)^3 + \dots$$

2. Simplify the expansion

This gives $8192 + 6656x + 2496x^2 + 572x^3 + \dots$.

3. Match the value

Since $2.0125 = 2 + 0.0125 = 2 + \frac{0.1}{8}$, use $x = 0.1$ in the expansion.

4. Calculate the approximation

$$8192 + 6656(0.1) + 2496(0.1)^2 + 572(0.1)^3 = 8883.132.$$

5. Decide the direction

The next terms in the expansion are positive because $x = 0.1 > 0$, so stopping after the x^3 term gives an underestimate.

Final answer

(a) $8192 + 6656x + 2496x^2 + 572x^3$. (b) 8883.132. (c)

Underestimate.

Question 5

Integration

5.

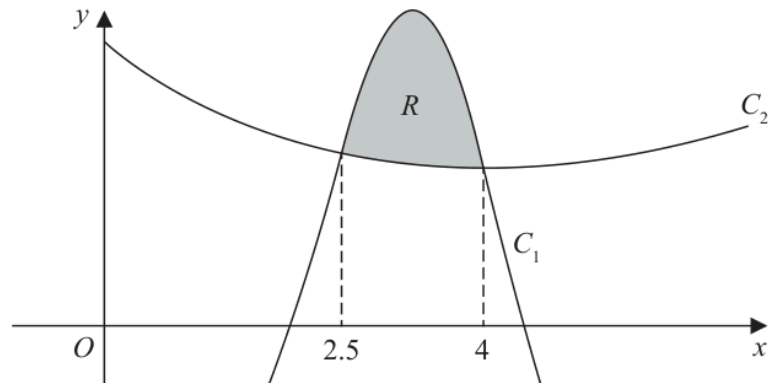


Figure 2

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

- (a) find, to 2 decimal places, an estimate for the area bounded by the curve C_1 , the line with equation $x = 2.5$, the x -axis and the line with equation $x = 4$

(4)

The curve C_2 has equation

$$y = x^{\frac{3}{2}} - 3x + 9 \quad x > 0$$

- (b) Find $\int \left(x^{\frac{3}{2}} - 3x + 9 \right) dx$

(3)

The region R , shown shaded in Figure 2, is bounded by the curves C_1 and C_2

- (c) Use the answers to part (a) and part (b) to find, to one decimal place, an estimate for the area of the region R .

(3)

Worked Solution - Question 5

1. Identify the strip width

The x values increase by 0.25, so $h = 0.25$.

2. Apply the trapezium rule

The area under C_1 is approximately

$$\frac{0.25}{2} \{5.453 + 5 + 2(7.764 + 9.375 + 9.964 + 9.367 + 7.626)\}.$$

3. Calculate the estimate

This gives 12.330625, so the estimate is 12.33 to 2 decimal places.

4. Integrate C two

$$\int (x^{3/2} - 3x + 9) dx = \frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x + c.$$

5. Find the area under C two

Between $x = 2.5$ and $x = 4$, the exact integral of C_2 is

$$\left[\frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x \right]_{2.5}^4 = 7.722 \dots$$

6. Find the shaded area

The shaded region is between C_1 and C_2 , so its area is approximately

$$12.330625 - 7.722 \dots = 4.608 \dots$$

7. Round the result

To one decimal place, the area of region R is 4.6.

Final answer

(a) 12.33. (b) $\frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x + c$. (c) 4.6.

Question 6

Circles

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

- (a) find the value of k .

(3)

The circle meets the coordinate axes at the points R , S and T .

- (b) Find the exact area of the triangle RST .

(4)

Worked Solution - Question 6

1. Complete the square

$$x^2 - 6x = (x - 3)^2 - 9 \text{ and } y^2 + 8y = (y + 4)^2 - 16.$$

2. Write centre-radius form

$$\text{The circle becomes } (x - 3)^2 + (y + 4)^2 = 25 - k.$$

3. Use the tangent condition

The centre is $(3, -4)$. Since the x -axis is tangent, the radius is the distance from the centre to the x -axis, which is 4.

4. Find k

$$\text{Thus } 25 - k = 4^2 = 16, \text{ so } k = 9.$$

5. Find the intercepts

With $k = 9$, the circle is $(x - 3)^2 + (y + 4)^2 = 16$. On the x -axis, $y = 0$, giving $x = 3$.

6. Find the y -axis points

$$\text{On the } y\text{-axis, } x = 0, \text{ so } 9 + (y + 4)^2 = 16. \text{ Hence } y = -4 \pm \sqrt{7}.$$

7. Calculate the triangle area

The vertical base on the y -axis has length $2\sqrt{7}$, and its perpendicular distance to $(3, 0)$ is 3. The area is $\frac{1}{2}(2\sqrt{7})(3) = 3\sqrt{7}$.

Final answer

(a) $k = 9$. (b) $3\sqrt{7}$.

Question 7

Laws of Logarithms

7. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0 \quad (4)$$

- (b) Show that -4 is a root of this cubic equation. (2)

- (c) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25) \quad (4)$$

Worked Solution - Question 7

Topic group

1. Use the power law

$$3 \log_3(2x - 1) = \log_3(2x - 1)^3.$$

2. Rewrite two as a logarithm

Since $2 = \log_3 9$, the right side is

$$\log_3 9 + \log_3(14x - 25) = \log_3\{9(14x - 25)\}.$$

3. Equate the arguments

$$\text{So } (2x - 1)^3 = 9(14x - 25).$$

4. Expand and simplify

$$8x^3 - 12x^2 + 6x - 1 = 126x - 225, \text{ which simplifies to} \\ 2x^3 - 3x^2 - 30x + 56 = 0.$$

5. Check the given root

Substituting $x = -4$ gives $2(-64) - 3(16) - 30(-4) + 56 = 0$, so -4 is a root.

6. Factorise the cubic

$$\text{Dividing by } (x + 4) \text{ gives } 2x^2 - 11x + 14 = (2x - 7)(x - 2).$$

7. Apply the logarithm domain

The possible roots are -4 , 2 , and $\frac{7}{2}$. The original logarithms require $2x - 1 > 0$ and $14x - 25 > 0$, so $x > \frac{25}{14}$.

8. State the valid solutions

Therefore the solutions are $x = 2$ and $x = \frac{7}{2}$.

Final answer

(a) Proven.

(b) -4 is a root of the cubic.

(c) $x = 2, \frac{7}{2}$.

Question 8

Trigonometric Equations

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta < 360^\circ$, the equation

$$3 \sin(\theta + 30^\circ) = 7 \cos(\theta + 30^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x (a \cos^2 x + b \cos x + c) = 0$$

where a , b and c are constants to be found.

- (b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

(6)

Worked Solution - Question 8

1. Solve for theta plus thirty

Let $u = \theta + 30^\circ$. Then $3 \sin u = 7 \cos u$, so $\tan u = \frac{7}{3}$.

2. Find u in the interval

Since $0 < \theta < 360^\circ$, $30^\circ < u < 390^\circ$. Thus $u = 66.801\dots^\circ$ or $246.801\dots^\circ$.

3. Return to theta

$\theta = u - 30^\circ$, giving $\theta = 36.8^\circ$ and 216.8° to one decimal place.

4. Rearrange the second equation

$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$ gives $3 \sin^3 x - 5 \sin x + 7 \sin x \cos x = 0$.

5. Factor out sine

$\sin x(3 \sin^2 x - 5 + 7 \cos x) = 0$.

6. Use sin squared

Since $\sin^2 x = 1 - \cos^2 x$, this becomes $\sin x(-3 \cos^2 x + 7 \cos x - 2) = 0$, or $\sin x(3 \cos^2 x - 7 \cos x + 2) = 0$.

7. Solve in the interval

For $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, $\sin x = 0$ gives $x = 0$. Also $3 \cos^2 x - 7 \cos x + 2 = 0$ gives $(3 \cos x - 1)(\cos x - 2) = 0$.

8. State the solutions

$\cos x = 2$ is impossible, so $\cos x = \frac{1}{3}$. The solutions are $x = -\cos^{-1}\left(\frac{1}{3}\right)$, 0, and $\cos^{-1}\left(\frac{1}{3}\right)$.

Final answer

$$(i) \theta = 36.8^\circ, 216.8^\circ. \quad (ii)(a) \ a = 3, \ b = -7, \ c = 2. \quad (ii)(b) \ x = -\cos^{-1}\left(\frac{1}{3}\right), 0, \cos^{-1}\left(\frac{1}{3}\right)$$

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Question 9

Applications of Differentiation

9.

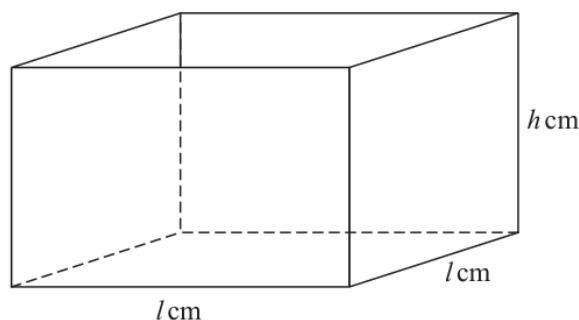


Figure 3

Figure 3 shows a sketch of a square based, open top box.

The height of the box is h cm, and the base edges each have length l cm.

Given that the volume of the box is $250\,000\text{ cm}^3$

(a) show that the external surface area, $S\text{ cm}^2$, of the box is given by

$$S = \frac{250\,000}{h} + 2000\sqrt{h} \quad (3)$$

(b) Use algebraic differentiation to show that S has a stationary point when $h = 250^k$ where k is a rational constant to be found. (5)

(c) Justify by further differentiation that this value of h gives the minimum external surface area of the box. (2)

(Total 10 marks)

Worked Solution - Question 9

Topic group

1. Use the volume

The volume is $l^2h = 250000$, so $l^2 = \frac{250000}{h}$.

2. Write the surface area

The open box has one square base and four side faces, so $S = l^2 + 4lh$.

3. Substitute l

Since $l = \sqrt{\frac{250000}{h}} = \frac{500}{\sqrt{h}}$, $4lh = 4 \cdot \frac{500}{\sqrt{h}} \cdot h = 2000\sqrt{h}$.

4. Show S

Therefore $S = \frac{250000}{h} + 2000\sqrt{h}$.

5. Differentiate S

$S = 250000h^{-1} + 2000h^{1/2}$, so $\frac{dS}{dh} = -250000h^{-2} + 1000h^{-1/2}$.

6. Set the derivative to zero

$-250000h^{-2} + 1000h^{-1/2} = 0$, so $1000h^{-1/2} = 250000h^{-2}$.

7. Solve for h

Multiplying by h^2 gives $1000h^{3/2} = 250000$, so $h^{3/2} = 250$ and $h = 250^{2/3}$.

8. Find k

Since $h = 250^k$, we have $k = \frac{2}{3}$.

9. Use the second derivative

$$\frac{d^2 S}{dh^2} = 500000h^{-3} - 500h^{-3/2}.$$

10. Check the sign

At $h = 250^{2/3}$, $h^{3/2} = 250$, so the second derivative is

$$\frac{500000}{250^2} - \frac{500}{250} = 8 - 2 = 6 > 0.$$

11. Conclude

The stationary point is therefore a minimum.

Final answer

(a) Proven: $S = \frac{250000}{h} + 2000\sqrt{h}$.

(b) $h = 250^{2/3}$, so $k = \frac{2}{3}$.

(c) Minimum.